

A Practical Guide to Speech recognition

TOPIC -- SPEECH RECOGNITION

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Neural networks became interesting in the late 1980s before beginning to dominate in the 2010s. Neural networks have been used in many aspects of speech recognition, such as phoneme classification, phoneme classification through multi-objective evolutionary algorithms, isolated word recognition, audiovisual speech recognition, audiovisual speaker recognition, and speaker adaptation. Neural networks make fewer explicit assumptions about feature statistical properties than HMMs. When used to estimate the probabilities of a speech segment, neural networks allow natural and efficient discriminative training. However, in spite of their effectiveness in classifying short-time units such as individual phonemes and isolated words, early neural networks were rarely successful for continuous recognition because of their limited ability to model temporal dependencies. One approach was to use neural networks for feature transformation, or dimensionality reduction. However, more recently, LSTM and related recurrent neural networks (RNNs), Time Delay Neural Networks (TDNN's), and transformers demonstrated improved performance.

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Acoustic signals are structured into a hierarchy of units, e.g. phonemes, words, phrases, and sentences; Each level provides additional constraints; e.g., known word pronunciations or legal word sequences, which can compensate for errors or uncertainties at a lower level; This hierarchy of constraints improves accuracy. By combining decisions probabilistically at all lower levels, and making ultimate decisions only at the highest level, speech recognition is broken into several phases. Computationally, it is a problem in which a sound pattern has to be recognized or classified into a category that represents a meaning to a human. Every acoustic signal can be broken into smaller sub-signals. As the more complex sound signal is divided, different levels are created, where at the top level are complex sounds made of simpler sounds on the lower level, etc. At the lowest level, simple and more probabilistic rules apply. These sounds are put together into more complex sounds on upper level, a new set of more deterministic rules predicts what the complex sound represents. The upper level of a deterministic rule should figure out the meaning of complex expressions. In order to expand our knowledge about speech recognition, we need to take into consideration neural networks. Neural network approaches use the following

steps:

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Software Sphinx toolkit is one starting point for experimenting with speech recognition. HTK book and accompanying toolkit Kaldi toolkit can be used. Common Voice (uses TensorFlow). Coqui STT (derived from Common Voice, using the same open-source license) Gboard supports speech recognition on all Android applications. Speech recognition is available in Microsoft Windows operating systems. Commercial cloud based speech recognition APIs are broadly available.

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Dynamic time warping was historically used for speech recognition, but was later displaced by HMM. Dynamic time warping measures similarity between two sequences that may vary in time or speed. For instance, similarities in walking patterns could be detected, even if in one video a person was walking slowly and in another was walking more quickly, or even if accelerations and decelerations came during one observation. DTW has been applied to video, audio, and graphics – any data that can be turned into a linear representation can be analyzed with DTW. This could handle speech at different speaking speeds. In general, it allows an optimal match between two sequences (e.g., time series) with certain restrictions. The sequences are "warped" non-linearly to match each other. This sequence alignment method is often used in the context of HMMs.

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Digitize the speech – for telephone speech, 8000 samples per second are captured; Compute features of spectral-domain of the speech (with Fourier transform); computed every 10ms, with one 10ms section called a frame; Sound is produced by air (or some other medium) vibration. Sound creates a wave that has two measures: amplitude (strength), and frequency (vibrations per second). Accuracy can be computed with the help of WER, which is calculated by aligning the recognized word and referenced word using dynamic string alignment. The problem may occur while computing the WER due to the difference between the sequence lengths of the recognized word and referenced word. The formula to compute the word error rate (WER) is:

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2010s By early the 2010s, speech recognition was differentiated from speaker recognition, and speaker independence was considered a major breakthrough. Until then, systems required a "training" period for each voice. In 2017, Microsoft

researchers reached the human parity milestone of transcribing conversational speech on the widely benchmarked Switchboard task. Multiple deep learning models were used to improve accuracy. The error rate was reported to be as low as 4 professional human transcribers working together on the same benchmark.

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Attention-based models Attention-based ASR models were introduced by Chan et al. of Carnegie Mellon University and Google Brain, and Bahdanau et al. of the University of Montreal in 2016. The model named "Listen, Attend and Spell" (LAS), literally "listens" to the acoustic signal, pays "attention" to all parts of the signal and "spells" out the transcript one character at a time. Unlike CTC-based models, attention-based models require conditional-independence assumptions and can learn all the components of a speech recognizer directly. This means that during deployment, no a priori language model is required, making it less demanding for applications with limited memory. Attention-based models immediately outperformed CTC models (with or without an external language model) and continued improving. Latent Sequence Decomposition (LSD) was proposed by Carnegie Mellon University, MIT, and Google Brain to directly emit sub-word units that are more natural than English characters. The University of Oxford and Google DeepMind extended LAS to "Watch, Listen, Attend and Spell" (WLAS) to handle lip reading and surpassed human-level performance.

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Speech recognition systems are based on HMMs. These are statistical models that output a sequence of symbols or quantities. HMMs are used in speech recognition because a speech signal can be viewed as a piecewise stationary signal or a short-time stationary signal. In a short time scale (e.g. 10 milliseconds), speech can be approximated as a stationary process. Speech can be thought of as a Markov model for many stochastic purposes. HMMs are popular because they can be trained automatically and are simple and computationally feasible. An HMM outputs a sequence of n -dimensional real-valued vectors (where n is an integer such as 10), outputting one every 10 milliseconds. The vectors consist of cepstral coefficients, obtained by a Fourier transform of a short window of speech and decorrelating the spectrum using a cosine transform, then taking the first (most significant) coefficients. The HMM tends to have, in each state, a statistical distribution that is a mixture of diagonal covariance Gaussians, which give a likelihood for each observed vector. Each word, or (for more general speech recognition systems), each phoneme, has a different output distribution; an HMM for a sequence of words or phonemes is made by concatenating the individual trained HMMs for the separate words and

phonemes. Speech recognition systems use combinations of standard techniques to improve results. A typical large-vocabulary system applies context dependency for the phonemes (so that phonemes with different left and right context have different realizations as HMM states). It uses cepstral normalization to handle speaker and recording conditions. It might use vocal tract length normalization (VTLN) for male-female normalization and maximum likelihood linear regression (MLLR) for more general adaptation. The features use delta and delta-delta coefficients to capture speech dynamics, and in addition might use heteroscedastic linear discriminant analysis (HLDA); or might use splicing and LDA-based projection, followed by HLDA or a global semi-tied covariance transform (also known as maximum likelihood linear transform (MLLT)). Many systems use discriminative training techniques that dispense with a purely statistical approach to HMM parameter estimation and instead optimize some classification-related measure of the training data. Examples are maximum mutual information (MMI), minimum classification error (MCE), and minimum phone error (MPE).